



Configuring VLANs for Network Isolation

Difficulty (out of 3 stars): ★★★

Recommended study time: 1 hour

Contents

1 About This Lab	2
1.1 Task Description	2
1.2 Objectives.....	2
2 Task Preparation	2
2.1 Network Topology	2
2.2 Initial Configuration.....	3
3 Task Implementation	4
3.1 Checking Basic Configurations	4
3.2 Configuring IP Addresses on S3 and S4.....	5
3.3 Creating VLANs on S1 and S2.....	6
3.4 Configuring Interface-based VLAN Assignment.....	6
3.5 Configuring Trunk Interfaces on S1 and S2	6
3.6 Configuring Hybrid Interfaces on S2	7
3.7 Configuring MAC Address-based VLAN Assignment.....	7
3.8 Checking Configurations	8
4 Verification	10
5 Quiz	10



1 About This Lab

1.1 Task Description

Ethernet technology allows data communication over shared media through Carrier Sense Multiple Access/Collision Detection (CSMA/CD). When an Ethernet network has a large number of hosts, collision becomes a serious problem and can lead to broadcast storms. This can degrade network performance or even result in a complete breakdown. Using switches to connect LANs can mitigate collisions, but broadcast may still pose an issue.

To alleviate broadcast storms, VLAN technology divides a physical LAN into multiple VLANs. Each VLAN functions as a separate broadcast domain, with hosts in the same VLAN able to directly communicate with one another, while those in different VLANs cannot. As a result, broadcast packets are confined within a single VLAN.

After completing this task, you will be able to configure Huawei switches, learn about Ethernet and VLAN technologies, and familiarize yourself with VLAN configurations.

1.2 Objectives

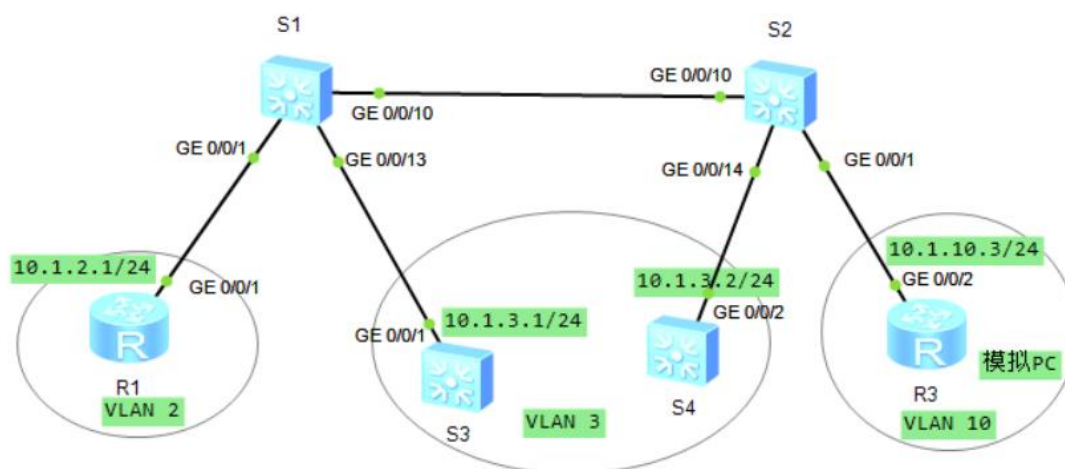
On completing this task, you will be able to:

1. Learn how to create a VLAN.
2. Learn how to configure access, trunk, and hybrid interfaces.
3. Learn how to configure VLANs based on interfaces.
4. Learn how to configure VLANs based on MAC addresses.
5. Learn how to view the MAC address table and VLAN information.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology





A company needs to divide a Layer 2 network into multiple VLANs based on service requirements. In addition, VLAN 10 requires a higher level of security and only specified PCs are allowed to access the Internet through VLAN 10

To meet this requirement, user interfaces of identical services on S1 and S2 can be assigned to the same VLAN, and interfaces with specified MAC addresses on S2 can be assigned to a VLAN.

2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]st
[Huawei]sys
[Huawei]sysname R1
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.1.2.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]int g0/0/2
[R3-GigabitEthernet0/0/2]ip address 10.1.10.3 24
[R3-GigabitEthernet0/0/2]quit
[R3]
```

- Initial configuration of S1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]
```

- Initial configuration of S2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S2
[S2]
```

- Initial configuration of S3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S3
[S3]
```

- Initial configuration of S4

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S4
```



[S4]

3 Task Implementation

3.1 Checking Basic Configurations

Run the **display ip interface brief** command on R1 and R2 to check interface configurations.

```
<R1>display ip int brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
```

The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 2

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	10.1.2.1/24	up	up
GigabitEthernet0/0/2	unassigned	down	down
NULL0	unassigned	up	up(s)

```
<R1>
```

```
<R3>display ip int brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
```

The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 2

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	10.1.10.3/24	up	up
NULL0	unassigned	up	up(s)

```
<R3>
```



3.2 Configuring IP Addresses on S3 and S4

Configure IP addresses on S3 and S4.

- **Scenario 1:**

This scenario applies to the environment where S3 and S4 support interface mode switching from Layer 2 to Layer 3. The IP addresses of the physical interfaces are 10.1.3.1/24 and 10.1.3.2/24, respectively.

```
[S3]interface GigabitEthernet0/0/1
[S3-GigabitEthernet0/0/1]undo portswitch
```

The interface working mode is changed to Layer 3. The **undo portswitch** command changes the working mode of an Ethernet interface from Layer 2 mode to Layer 3 mode.

```
[S3-GigabitEthernet0/0/1]ip address 10.1.3.1 24
```

Configure S4 in the same way.

```
[S4]interface GigabitEthernet0/0/2
[S4-GigabitEthernet0/0/2]undo portswitch
[S4-GigabitEthernet0/0/2]ip address 10.1.3.2 24
```

- **Scenario 2:**

This scenario applies to the environment where S3 and S4 do not support interface mode switching from Layer 2 to Layer 3. Configure IP addresses on S3 and S4. Set the IP addresses of VLANIF3 to 10.1.3.1/24 and 10.1.3.2/24, respectively.

This task applies to scenario 2.

(1) Create VLAN 3 on S3 and S4.

```
[S3]vlan 3
[S3-vlan3]

[S4]vlan 3
[S4-vlan3]
```

(2) Configure interfaces on S3 and S4 as access interfaces and assign them to corresponding VLANs.

```
[S3]interface GigabitEthernet0/0/1
[S3-GigabitEthernet0/0/1]port link-type access
[S3-GigabitEthernet0/0/1]port default vlan 3
[S3-GigabitEthernet0/0/1]quit

[S4]interface GigabitEthernet0/0/2
[S4-GigabitEthernet0/0/2]port link-type access
[S4-GigabitEthernet0/0/2]port default vlan 3
[S4-GigabitEthernet0/0/2]quit
```

(3) # Create VLANIF interfaces and configure IP addresses.

```
[S3] interface Vlanif 3
```

The **interface vlanif *vlan-id*** command creates a VLANIF interface and displays the VLANIF interface view.

```
[S3-Vlanif3]ip address 10.1.3.1 24
```

```
[S4] interface Vlanif 3
```



```
[S4-Vlanif3]ip address 10.1.3.2 24
```

3.3 Creating VLANs on S1 and S2

Create VLANs 2, 3, and 10 on S1 and S2.

```
[S1]vlan batch 2 to 3 10
```

Info: This operation may take a few seconds. Please wait for a moment...done.

VLANs 2, 3, and 10 are created successfully in a batch. The **vlan *vlan-id*** command creates a VLAN and displays the VLAN view or displays the view of an existing VLAN. The **vlan batch { *vlan-id1* [to *vlan-id2*] }** command creates VLANs in a batch.

```
[S2]vlan batch 2 to 3 10
```

3.4 Configuring Interface-based VLAN Assignment

Configure user interfaces on S1 and S2 as access interfaces and assign them to corresponding VLANs.

```
[S1]interface GigabitEthernet0/0/1
```

```
[S1-GigabitEthernet0/0/1]port link-type access
```

The **port link-type { access | hybrid | trunk }** command specifies the link type of an interface, which can be access, trunk, or hybrid.

```
[S1-GigabitEthernet0/0/1]port default vlan 2
```

The **port default vlan *vlan-id*** command configures the default VLAN of an interface and assigns the interface to the VLAN.

```
[S1-GigabitEthernet0/0/1]quit
```

```
[S1]interface GigabitEthernet0/0/13
```

```
[S1-GigabitEthernet0/0/13]port link-type access
```

```
[S1-GigabitEthernet0/0/13]port default vlan 3
```

```
[S1-GigabitEthernet0/0/13]quit
```

```
[S2]interface GigabitEthernet0/0/14
```

```
[S2-GigabitEthernet0/0/14]port link-type access
```

```
[S2-GigabitEthernet0/0/14]port default vlan 3
```

```
[S2-GigabitEthernet0/0/14]quit
```

3.5 Configuring Trunk Interfaces on S1 and S2

Configure the interfaces connecting S1 and S2 as trunk interfaces and allow only packets from VLANs 2 and 3 to pass through.

```
[S1]interface GigabitEthernet0/0/10
```

```
[S1-GigabitEthernet0/0/10]port link-type trunk
```

```
[S1-GigabitEthernet0/0/10]port trunk allow-pass vlan 2 3
```

The **port trunk allow-pass vlan** command adds a trunk interface to the specified VLANs.

```
[S1-GigabitEthernet0/0/10]undo port trunk allow-pass vlan 1
```



The **undo port trunk allow-pass vlan** command deletes a trunk interface from the specified VLANs. By default, VLAN 1 is in the VLAN allowlist. If VLAN 1 is not used for any service, it needs to be deleted for security purposes.

```
[S2]interface GigabitEthernet0/0/10
[S2-GigabitEthernet0/0/10]port link-type trunk
[S2-GigabitEthernet0/0/10]port trunk allow-pass vlan 2 3
[S2-GigabitEthernet0/0/10]undo port trunk allow-pass vlan 1
```

3.6 Configuring Hybrid Interfaces on S2

Configure GigabitEthernet0/0/1, GigabitEthernet0/0/2, and GigabitEthernet0/0/3 on S2 as hybrid interfaces. Configure them to allow packets from VLANs assigned based on MAC addresses to pass through.

On access and trunk interfaces, MAC address-based VLAN assignment can be used only when the VLAN ID is the same as the PVID. Therefore, it is recommended that you configure MAC address-based VLAN assignment on a hybrid interface to receive untagged packets from multiple VLANs.

```
[S2]interface GigabitEthernet0/0/1
[S2-GigabitEthernet0/0/1]port link-type hybrid
[S2-GigabitEthernet0/0/1]port hybrid untagged vlan 10
```

The **port hybrid untagged vlan** command adds a hybrid interface to the specified VLANs. Frames of the VLANs then pass through the hybrid interface in untagged mode.

```
[S2-GigabitEthernet0/0/1]quit
[S2]interface GigabitEthernet0/0/2
[S2-GigabitEthernet0/0/2]port link-type hybrid
[S2-GigabitEthernet0/0/2]port hybrid untagged vlan 10
[S2-GigabitEthernet0/0/2]quit
[S2]interface GigabitEthernet0/0/3
[S2-GigabitEthernet0/0/3]port link-type hybrid
[S2-GigabitEthernet0/0/3]port hybrid untagged vlan 10
[S2-GigabitEthernet0/0/3]quit
```

3.7 Configuring MAC Address-based VLAN Assignment

As shown in the networking diagram, R3 simulates a special service PC. Assume that the MAC address of the PC is a008-6fe1-0c46. The PC is expected to connect to the network through any of GigabitEthernet0/0/1, GigabitEthernet0/0/2, and GigabitEthernet0/0/3 on S2 and transmit data through VLAN 10.

1. Configure S2 to associate the MAC address of the PC with VLAN 10.

MAC address-based VLAN assignment associates MAC addresses with VLANs, defines VLAN members based on the source MAC addresses of packets, and adds the VLAN tag to specified packets before sending the packets. This VLAN assignment method is independent of the location, providing a higher level of security and flexibility.

```
[S2] vlan 10
[S2-vlan10] mac-vlan mac-address a008-6fe1-0c46
```

The **mac-vlan mac-address** command associates a MAC address with a VLAN.



2. Configure the interfaces connecting S1 and S2 to allow packets from VLAN 10 to pass through.

The interfaces need to allow tagged frames from multiple VLANs to pass through. Therefore, the interfaces can be configured as trunk interfaces.

```
[S1]interface GigabitEthernet0/0/10
[S1-GigabitEthernet0/0/10]port trunk allow-pass vlan 10
[S1-GigabitEthernet0/0/10]quit
```

```
[S2]interface GigabitEthernet0/0/10
[S2-GigabitEthernet0/0/10]port trunk allow-pass vlan 10
[S2-GigabitEthernet0/0/10]quit
```

3. Configure S2 and enable MAC address-based VLAN assignment on GE0/0/1, GE0/0/2, and GE0/0/3. To enable an interface to forward packets based on associations between MAC addresses and VLANs, you must enable MAC address-based assignment on the interface.

```
[S2]interface GigabitEthernet0/0/1
[S2-GigabitEthernet0/0/1]mac-vlan enable
```

The **mac-vlan enable** command enables MAC address-based VLAN assignment on an interface.

```
[S2-GigabitEthernet0/0/1]quit
[S2]interface GigabitEthernet0/0/2
[S2-GigabitEthernet0/0/2]mac-vlan enable
[S2-GigabitEthernet0/0/2]quit
[S2]interface GigabitEthernet0/0/3
[S2-GigabitEthernet0/0/3]mac-vlan enable
[S2-GigabitEthernet0/0/3]quit
```

3.8 Checking Configurations

1. Check VLAN information on the switches.

The **display vlan** command displays information about VLANs. The **display vlan verbose** command displays detailed information about a specified VLAN, including the ID, type, description, and status of the VLAN, status of the traffic statistics collection function, interfaces in the VLAN, and mode in which the interfaces are assigned to the VLAN.

```
[S1]display vlan
```

The total number of vlans is : 4

```
-----
U: Up;           D: Down;           TG: Tagged;      UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
-----
```

VID	Type	Ports
1	common	UT: GE0/0/2(D) GE0/0/3(D) GE0/0/4(D) GE0/0/5(D) GE0/0/6(D) GE0/0/7(D) GE0/0/8(D) GE0/0/9(D) GE0/0/11(D) GE0/0/12(D) GE0/0/14(D) GE0/0/15(D) GE0/0/16(D) GE0/0/17(D) GE0/0/18(D) GE0/0/19(D) GE0/0/20(D) GE0/0/21(D) GE0/0/22(D) GE0/0/23(D) GE0/0/24(D)
2	common	UT: GE0/0/1(U)



```

3    common  TG: GE0/0/10(U)
      UT: GE0/0/13(U)
10   common  TG: GE0/0/10(U)

```

VID	Status	Property	MAC-LRN Statistics		Description
1	enable	default	enable	disable	VLAN 0001
2	enable	default	enable	disable	VLAN 0002
3	enable	default	enable	disable	VLAN 0003
10	enable	default	enable	disable	VLAN 0010

```

[S2]display vlan
The total number of vlans is : 4

```

```

-----
U: Up;          D: Down;          TG: Tagged;      UT: Untagged;
MP: Vlan-mapping;  ST: Vlan-stacking;
#: ProtocolTransparent-vlan;  *: Management-vlan;
-----

```

VID	Type	Ports			
1	common	UT: GE0/0/1(U)	GE0/0/2(D)	GE0/0/3(D)	GE0/0/4(D)
		GE0/0/5(D)	GE0/0/6(D)	GE0/0/7(D)	GE0/0/8(D)
		GE0/0/9(D)	GE0/0/11(D)	GE0/0/12(D)	GE0/0/13(D)
		GE0/0/15(D)	GE0/0/16(D)	GE0/0/17(D)	GE0/0/18(D)
		GE0/0/19(D)	GE0/0/20(D)	GE0/0/21(D)	GE0/0/22(D)
		GE0/0/23(D)	GE0/0/24(D)		
2	common	TG: GE0/0/10(U)			
3	common	UT: GE0/0/14(U)			
		TG: GE0/0/10(U)			
10	common	UT: GE0/0/1(U)	GE0/0/2(D)	GE0/0/3(D)	
		TG: GE0/0/10(U)			

VID	Status	Property	MAC-LRN Statistics		Description
1	enable	default	enable	disable	VLAN 0001
2	enable	default	enable	disable	VLAN 0002
3	enable	default	enable	disable	VLAN 0003
10	enable	default	enable	disable	VLAN 0010

2. Check the MAC address-based VLAN assignment information on the switch.

```

[S2]display mac-vlan vlan 10

```

MAC Address	MASK	VLAN	Priority
00e0-fc1c-47a7	ffff-ffff-ffff	10	0

```

Total MAC VLAN address count: 1

```

The **display mac-vlan** command displays the configuration of MAC address-based VLAN assignment.



4 Verification

Test the device connectivity and verify the VLAN configuration.

1. Ping S4 from S3. Check that the ping operation is successful.
2. Ping other devices from R1. Check that the ping operations fail.
3. Run the **display mac-address verbose** command on S1 and S2 to check the MAC address tables on the switches.

5 Quiz

1. A Layer 2 Ethernet switch generates a MAC address table based on the (A) of the packets received by the interface.
 - A. Source MAC addresses
 - B. Destination MAC addresses
 - C. Source IP addresses
 - D. Destination IP addresses
2. A switch has eight interfaces. If a unicast frame enters the switch through one of the interfaces but the switch cannot find the MAC address entry corresponding to the destination MAC address of the frame, what will the switch do? (B)
 - A. Discard the frame.
 - B. Flood the frame.
 - C. Forward the frame in P2P mode.
3. (Multiple-answer question) Which of the following statements about VLAN are incorrect? (AC)
 - A. VLAN technology can isolate a large collision domain into several small collision domains.
 - B. VLAN technology can isolate a large Layer 2 broadcast domain into several small Layer 2 broadcast domains.
 - C. PCs in different VLANs cannot communicate.
 - D. PCs in the same VLAN can communicate at Layer 2.
4. If the PVID of a trunk interface is 5 and the **port trunk allow-pass vlan 2 3** command is run on the interface, which VLANs' frames can be transmitted through the trunk interface?

Answer: After the **port trunk allow-pass vlan 2 3** command is run, the frames of VLAN 5 cannot be transmitted through the trunk interface. By default, the frames of VLAN 1 can be transmitted through the trunk interface. Therefore, the frames of VLANs 1 through 3 can all be transmitted through the trunk interface.